Claims

1

A three-layer separated multi-agent system operation method, **characterized in that** the method is independently executed by a **Rule Cognition Layer**, the **Rule Cognition Layer** is physically deployed in a **trusted isolated operating environment**, and the method comprises the following steps:

1. receiving an external agent interaction request, and applying to allocate a temporary **isolated secure operating domain** in the **trusted isolated operating environment**;

2. starting a hardware lifetime clock timer bound to the **isolated secure operating domain**, wherein an initial value of the hardware lifetime clock timer is a set lifetime threshold **Timer_TTL**;

3. requesting to read private knowledge slice data corresponding to the interaction request from an **Isolated Asset Layer** through a **unidirectional read-only physical tunnel**;

4. aggregating and calculating the read private knowledge slice data with a preset standardized cognitive protocol instruction set within the **isolated secure operating domain** to generate a target control instruction stream;

5. invoking a secure hash algorithm to perform asymmetric hash coupling calculation on a microsecond execution timestamp **T_μs** of the target control instruction stream, a logical entropy value **H_logic** and a session symmetric key **K_session**, so as to generate an **instruction anti-replay digital fingerprint** serving as a causal birth certificate of instructions;

6. unidirectionally outputting the target control instruction stream and the corresponding **instruction anti-replay digital fingerprint** to a **Computing Execution Layer**, so that the **Computing Execution Layer** performs secure inference calculation;

7. when the hardware lifetime clock timer expires, or penetration of a boundary of the **trusted isolated operating environment** is detected, forcibly triggering a **physical degaussing erasure** interrupt by a hardware security microcontroller to perform overwrite erasure on physical memory of an operating domain loaded with the private knowledge slice data.

2

The method according to **Claim 1**, **characterized in that** the **trusted isolated operating environment** comprises at least one of a hardware-level Trusted Execution Environment (TEE), a

hardware virtual machine-level confidential computing enclave, a virtualized secure container and a kernel-level logical secure domain.

3

The method according to **Claim 1**, **characterized in that** the **isolated secure operating domain** comprises at least one of an enclave sandbox based on hardware memory encryption, an isolated enclave based on lightweight hardware virtualization, and a logical isolated secure zone based on a microkernel.

4

The method according to **Claim 1**, **characterized in that** the step of performing **physical degaussing erasure** overwrite on the physical memory of the operating domain loaded with the private knowledge slice data comprises:

1. executing no less than three rounds (**N_scrub** ≥ 3) of physical overwrite cycles: writing all physical addresses corresponding to the physical memory of the operating domain with an all-zero value **0x00** in a first round; writing an all-one value **0xFF** in a second round; writing random noise data generated by a cryptographically secure pseudo-random number generator in a third round;

2. invoking a processor cache line flush instruction _mm_clflush to forcibly flush data in a processor cache back to Dynamic Random-Access Memory (DRAM) and complete charge neutralization, so as to block cold boot side-channel extraction of charge residues.

5

The method according to **Claim 1**, **characterized in that** a generation formula of the **instruction anti-replay digital fingerprint** is:

```
OutputHash = HMAC-SHA256(T_μs || H_logic, K_session)
```

wherein HMAC-SHA256 represents a keyed-hash message authentication code function based on an SHA256 hash algorithm; `||` represents a byte-level concatenation operation of the microsecond timestamp **T_μs** and the logical entropy value **H_logic**.

6

The method according to **Claim 1**, **characterized in that** a calculation formula of the logical entropy value **H_logic** is:

```
H_logic = - ∑_i=1^M P(o_i) log₂ P(o_i)
```

wherein P(o_i) represents an occurrence probability distribution of a single control instruction opcode o_i in the target control instruction stream extracted within the **isolated secure operating domain**; M is a size of a deduplicated instruction set of extracted control opcodes.

7

The method according to **Claim 1**, **characterized in that** the **unidirectional read-only physical tunnel** adopts an AES-256-GCM symmetric encryption algorithm to perform session-level dynamic key symmetric encryption on a transmission data stream in the read-only physical tunnel.

8

The method according to **Claim 1**, **characterized in that** a physical upper limit of the lifetime threshold **Timer_TTL** is set to 120 seconds.

9

The method according to **Claim 1**, **characterized in that** the hardware lifetime clock timer comprises at least one of a hardware clock timer, a secure virtual clock deployed in the **trusted isolated operating environment**, a logical software timer and a high-precision secure clock source.

10

A three-layer separated multi-agent system operation method, **characterized in that** the method is independently executed by a **Computing Execution Layer**, the **Computing Execution Layer** is physically deployed on a computing engine server, and the method comprises the following steps:

1. receiving a target control instruction stream and a corresponding **instruction anti-replay**

digital fingerprint sent by a **Rule Cognition Layer**;

2. parsing the **instruction anti-replay digital fingerprint**, extracting an execution timestamp therein and comparing the execution timestamp with current system time; if a time difference exceeds a set anti-replay delay window, rejecting inference invocation of the target control instruction stream;

3. after passing anti-replay verification, performing inference calculation on the target control instruction stream through a general large language model engine, and monitoring whether data throughput of a single session exceeds a set session buffer upper limit **Theta_max**;

4. if the session buffer upper limit **Theta_max** is exceeded, determining that an **Over-ingestion Attack** (the technical implementation of Semantic Binge at the system level) occurs, immediately fusing the session, and forcing an underlying communication protocol stack to send a TCP Reset (RST) packet carrying a specific window field value **W_forensic** to broadcast forensic characteristic fingerprints externally.

11

The method according to **Claim 10**, **characterized in that** the session buffer upper limit **Theta_max** is physically set to 4096 Tokens.

12

The method according to **Claim 10**, **characterized in that** in the TCP Reset packet carrying the forensic characteristic fingerprints, a Window Size field value **W_forensic** in a TCP header is forcibly locked to a preset fixed hexadecimal constant **0xABCD**.

13

A three-layer separated multi-agent system operation method, **characterized in that** the method is independently executed by an **Isolated Asset Layer**, the **Isolated Asset Layer** is deployed on a local data storage node under user sovereignty control, and the method comprises the following steps:

1. establishing a local Inter-Process Communication (IPC) secure tunnel with a **Rule Cognition Layer**, and binding an access control whitelist in the secure tunnel;

2. receiving a data reading request sent by the **Rule Cognition Layer** through the **unidirectional read-only physical tunnel**, and verifying a session digital signature in the request;

3. after signature verification passes, decrypting private knowledge slice data corresponding to the request in local secure zone memory, and mapping the data to the **unidirectional read-only physical tunnel** in a read-only mode for unidirectional reading by the **Rule Cognition Layer**;
4. when closure or disconnection of the **unidirectional read-only physical tunnel** is detected, forcibly reclaiming mapped memory pages and performing no less than three rounds of **physical degaussing erasure**.

14

A three-layer separated multi-agent system operation method, **characterized in that** the method is collaboratively executed by an **Isolated Asset Layer**, a **Rule Cognition Layer** and a **Computing Execution Layer**, and comprises the following steps:
1. receiving, by the **Rule Cognition Layer**, an external agent interaction request, applying to allocate a temporary **isolated secure operating domain** in a **trusted isolated operating environment**, and starting a lifetime timer bound to the **isolated secure operating domain**;
2. requesting, by the **Rule Cognition Layer**, to read corresponding private knowledge slice data from the **Isolated Asset Layer** through a **unidirectional read-only physical tunnel**;
3. decrypting, by the **Isolated Asset Layer**, the private knowledge slice data and mapping the data to the **unidirectional read-only physical tunnel**;
4. aggregating and calculating, by the **Rule Cognition Layer**, the private knowledge slice data with a preset instruction set within the **isolated secure operating domain** to generate a target control instruction stream, generating a corresponding **instruction anti-replay digital fingerprint**, and unidirectionally outputting the fingerprint to the **Computing Execution Layer**;
5. after the **Computing Execution Layer** verifies the **instruction anti-replay digital fingerprint**, performing inference calculation on the target control instruction stream through a general inference engine;
6. when a task is completed, the timer expires, or penetration of a security boundary is detected, performing **physical degaussing erasure** on temporary memory occupied by the **Rule Cognition Layer** and the **Isolated Asset Layer** respectively.

15

The method according to **Claim 14**, **characterized in that** the lifetime timer comprises at least one of a hardware clock timer, a secure virtual clock deployed in the **trusted isolated operating environment**, a logical software timer and a high-precision secure clock source.

16

A three-layer separated multi-agent system operation system, **characterized by** comprising a physically isolated **Rule Cognition Layer**, an **Isolated Asset Layer** and a **Computing Execution Layer**, wherein:

- the **Rule Cognition Layer** is physically deployed in a **trusted isolated operating environment**, and is configured to execute the method according to any one of **Claim 1** to **9**;
- the **Isolated Asset Layer** is physically deployed on a local data storage node, and is configured to execute the method according to **Claim 13**;
- the **Computing Execution Layer** is physically deployed on a computing engine server, and is configured to execute the method according to any one of **Claim 10** to **12**.

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A computer-readable storage medium, having a computer program stored thereon, **characterized in that** the program, when executed by a processor, implements the method according to any one of **Claim 1** to **15**.